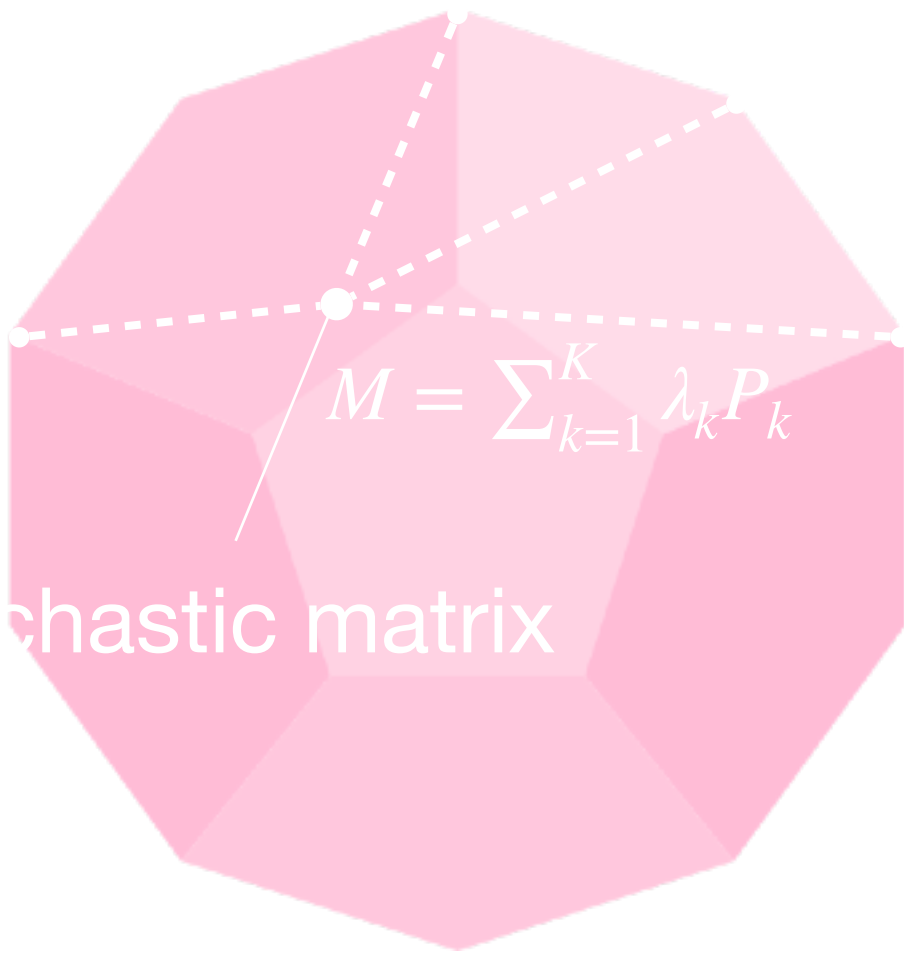


Not differentiable in general

adding regularization

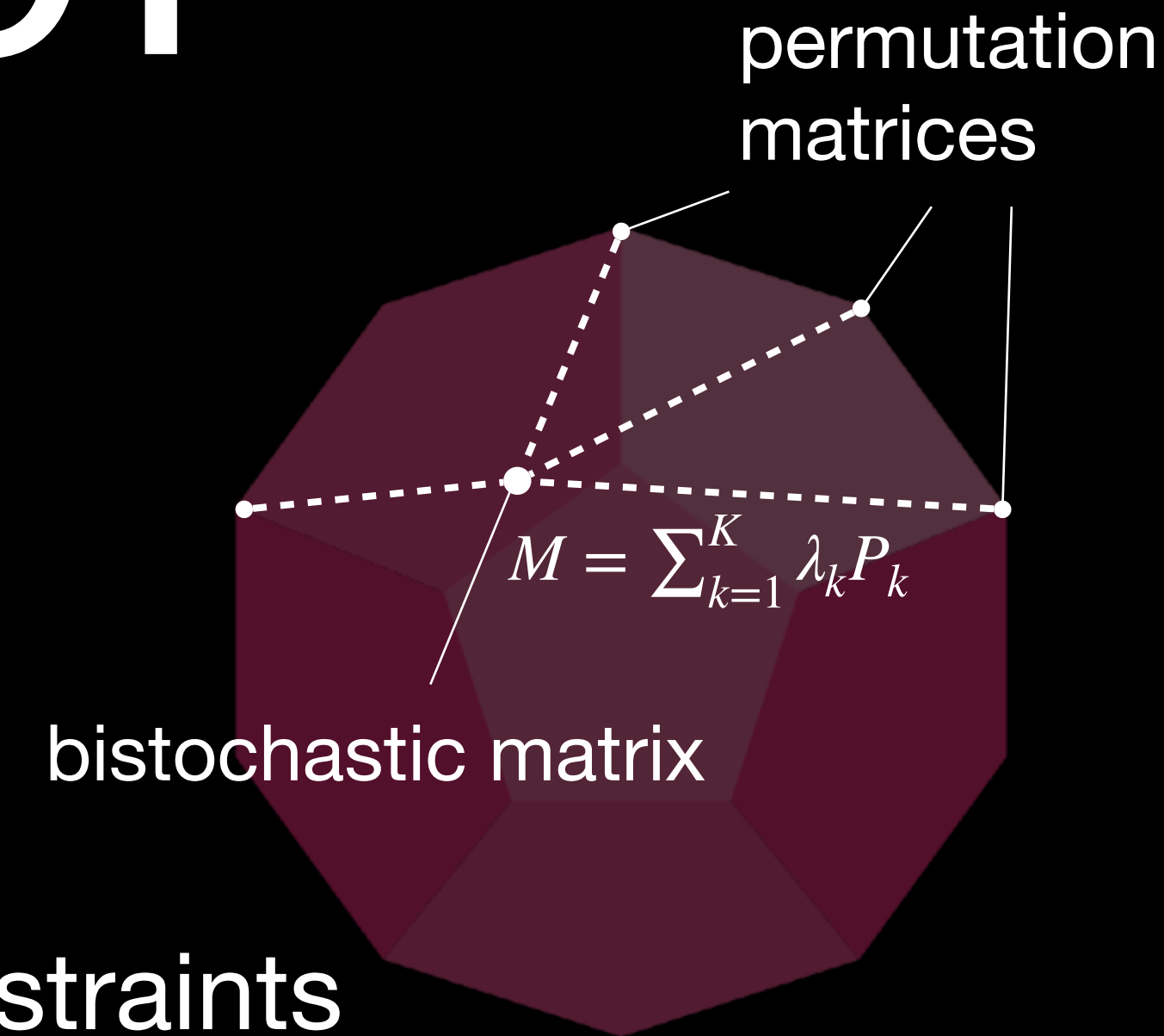


$$M = \sum_{k=1}^K \lambda_k P_k$$

stochastic matrix

Differentiability of OT

- **Recall:** classic discrete OT is solution to a linear program
- **Not differentiable in general:**
 - Optimal plan π^* may not be unique
 - Small perturbations in α, β can change active set of constraints
- Differentiability requires:
 - Uniqueness and stability of optimal coupling
 - Strict convexity of cost (e.g., squared distance)
- Often addressed by **adding regularization**



Differentiability of EOT

- Regularized formulation:



Kantorovich's Problem:

(Discrete version + entropy regularized)

$$\min_{\pi \in \Pi(\mathbf{a}, \mathbf{b})} \langle \pi, \mathbf{C} \rangle - \varepsilon H(\pi)$$